
Problem Set 2

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PY 451 Section A1

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Problem 1

i feel bad for whoever's grading this

1.

$$\langle b_1 | A | a_1 \rangle = (-1 \quad -i)^* \begin{pmatrix} 1 & 2i \\ -2 & i \end{pmatrix} \begin{pmatrix} 3i \\ 2 \end{pmatrix} = (-1 \quad i) \begin{pmatrix} 3i + 4i \\ -6i + 2i \end{pmatrix} = -7i + 4.$$

2.

$$\begin{aligned} \langle b_1 | A^\dagger | a_1 \rangle &= (-1 \quad -i)^* \begin{pmatrix} 1 & -2 \\ 2i & i \end{pmatrix}^* \begin{pmatrix} 3i \\ 2 \end{pmatrix} = (-1 \quad i) \begin{pmatrix} 1 & -2 \\ -2i & -i \end{pmatrix} \begin{pmatrix} 3i \\ 2 \end{pmatrix} = (-1 \quad i) \begin{pmatrix} 3i - 4 \\ -6 - 2i \end{pmatrix} \\ &= (4 - 3i) + (2 + 6i) = 6 + 3i. \end{aligned}$$

3.

$$(\langle b_1 | A | a_1 \rangle)^\dagger = (-7i + 4)^* = 4 + 7i.$$

4.

$$A | a_1 \rangle = \begin{pmatrix} 1 & 2i \\ -2 & i \end{pmatrix} \begin{pmatrix} 3i \\ 2 \end{pmatrix} = 3i \begin{pmatrix} 1 \\ -2 \end{pmatrix} + 2 \begin{pmatrix} 2i \\ i \end{pmatrix} = \begin{pmatrix} 7i \\ -4i \end{pmatrix}.$$

5.

$$\langle b_1 | A = (-1 \quad -i)^* \begin{pmatrix} 1 & 2i \\ -2 & i \end{pmatrix} = (-1 \quad i) \begin{pmatrix} 1 & 2i \\ -2 & i \end{pmatrix} = (-1 - 2i \quad -2i - 1).$$

6.

$$\langle b_1 | a_1 \rangle = (-1 \quad -i)^* \begin{pmatrix} 3i \\ 2 \end{pmatrix} = (-1 \quad i) \begin{pmatrix} 3i \\ 2 \end{pmatrix} = -3i + 2i = -i.$$

7.

$$\langle a_1 | b_1 \rangle = (\langle b_1 | a_1 \rangle)^* = (-i)^* = i.$$

8.

$$|b_1\rangle\langle a_1| = \begin{pmatrix} -1 \\ -i \end{pmatrix} (3i \quad 2)^* = \begin{pmatrix} -1 \\ -i \end{pmatrix} (-3i \quad 2) = \begin{pmatrix} 3i & -2 \\ -3 & -2i \end{pmatrix}.$$

9.

$$A | b_1 \rangle \langle a_1 | = \begin{pmatrix} 1 & 2i \\ -2 & i \end{pmatrix} \begin{pmatrix} 3i & -2 \\ -3 & -2i \end{pmatrix} = \begin{pmatrix} 3i - 6i & -2 + 4 \\ -6i - 3i & 4 + 2 \end{pmatrix} = \begin{pmatrix} -3i & 2 \\ -9i & 6 \end{pmatrix}.$$

10.

$$A | b_1 \rangle \langle a_1 | A = \begin{pmatrix} -3i & 2 \\ -9i & 6 \end{pmatrix} \begin{pmatrix} 1 & 2i \\ -2 & i \end{pmatrix} = \begin{pmatrix} -3i - 4 & 6 + 2i \\ -9i - 12 & 18 + 6i \end{pmatrix}.$$

11. why did we have to do this

$$A^2|b_1\rangle\langle a_1| = \begin{pmatrix} 1 & 2i \\ -2 & i \end{pmatrix} \begin{pmatrix} -3i & 2 \\ -9i & 6 \end{pmatrix} = \begin{pmatrix} -3i-18 & 2+12i \\ 6i+9 & -4+6i \end{pmatrix}.$$

12.

$$(|b_1\rangle\langle a_1|)^2 = \begin{pmatrix} 3i & -2 \\ -3 & -2i \end{pmatrix} \begin{pmatrix} 3i & -2 \\ -3 & -2i \end{pmatrix} = \begin{pmatrix} -9+6 & -6+4i \\ -9i+6i & 6-4 \end{pmatrix} = \begin{pmatrix} -3 & -2i \\ -3i & 2 \end{pmatrix}.$$

13.

$$A^\dagger|a_1\rangle = \begin{pmatrix} 1 & -2 \\ 2i & i \end{pmatrix}^* \begin{pmatrix} 3i \\ 2 \end{pmatrix} = \begin{pmatrix} 1 & -2 \\ -2i & -i \end{pmatrix} \begin{pmatrix} 3i \\ 2 \end{pmatrix} = \begin{pmatrix} 3i-4 \\ 6-2i \end{pmatrix}.$$

14. why do $|a_1\rangle$ and $|b_1\rangle$ have subscripts if we're only using a single a and b like we could just do $|a\rangle$ and $|b\rangle$ and it would be the same

$$(A|b_1\rangle)^\dagger = \left(\begin{pmatrix} 1 & 2i \\ -2 & i \end{pmatrix} \begin{pmatrix} -1 \\ -i \end{pmatrix} \right)^\dagger = \begin{pmatrix} -1+2 \\ 2+1 \end{pmatrix}^\dagger = \begin{pmatrix} 1 & 3 \end{pmatrix}.$$

15.

$$\begin{aligned} A|b_1\rangle - \lambda|b_1\rangle &= \begin{pmatrix} 1 & 2i \\ -2 & i \end{pmatrix} \begin{pmatrix} -1 \\ -i \end{pmatrix} - (2+3i) \begin{pmatrix} -1 \\ -i \end{pmatrix} = \begin{pmatrix} -1+2 \\ 2+1 \end{pmatrix} - \begin{pmatrix} -2-3i \\ -2i+3 \end{pmatrix} \\ &= \begin{pmatrix} 1 \\ 3 \end{pmatrix} + \begin{pmatrix} 2+3i \\ -3+2i \end{pmatrix} = (5+3i2i). \end{aligned}$$

16.

$$\lambda A = (2+3i) \begin{pmatrix} 1 & 2i \\ -2 & i \end{pmatrix} = \begin{pmatrix} 2+3i & 4i-6 \\ -4-6i & 2i-3 \end{pmatrix}.$$

17.

$$(\lambda - \langle a_1 | b_1 \rangle)A = (2+3i-i)A = (2+2i) \begin{pmatrix} 1 & 2i \\ -2 & i \end{pmatrix} = 4i-4 \begin{pmatrix} 2+2i & \\ -4-4i & 2i-2 \end{pmatrix}.$$

18.

$$\begin{aligned} (\langle a_1 | a_1 \rangle + \langle b_1 | b_1 \rangle)^{-1} &= \left((3i \ 2)^* \begin{pmatrix} 3i \\ 2 \end{pmatrix} + (-1 \ -i)^* \begin{pmatrix} -1 \\ -i \end{pmatrix} \right)^{-1} \\ &= \left((-3i \ 2) \begin{pmatrix} 3i \\ 2 \end{pmatrix} + (-1 \ i) \begin{pmatrix} -1 \\ -i \end{pmatrix} \right)^{-1} = (9+4+1+1)^{-1} = \frac{1}{15}. \end{aligned}$$

19.

$$\begin{aligned} (|a_1\rangle\langle a_1| + |b_1\rangle\langle b_1|)^{-1} &= \left(\begin{pmatrix} 3i \\ 2 \end{pmatrix} (-3i \ 2) + \begin{pmatrix} -1 \\ -i \end{pmatrix} (-1 \ i) \right)^{-1} = \left(\begin{pmatrix} 9 & 6i \\ -6i & 4 \end{pmatrix} + \begin{pmatrix} 1 & -i \\ i & 1 \end{pmatrix} \right)^{-1} \\ &= \begin{pmatrix} 10 & 5i \\ -5i & 5 \end{pmatrix}^{-1} = \frac{1}{50+25} \begin{pmatrix} 5 & -5i \\ 5i & 10 \end{pmatrix} = \frac{1}{15} \begin{pmatrix} 1 & -i \\ i & 2 \end{pmatrix}. \end{aligned}$$

20.

$$AA^\dagger = \begin{pmatrix} 1 & 2i \\ -2 & i \end{pmatrix} \begin{pmatrix} 1 & -2 \\ -2i & -i \end{pmatrix} = \begin{pmatrix} 1+4 & -2+2 \\ -2+2 & 4+1 \end{pmatrix} = \begin{pmatrix} 5 & \\ & 5 \end{pmatrix}.$$

21.

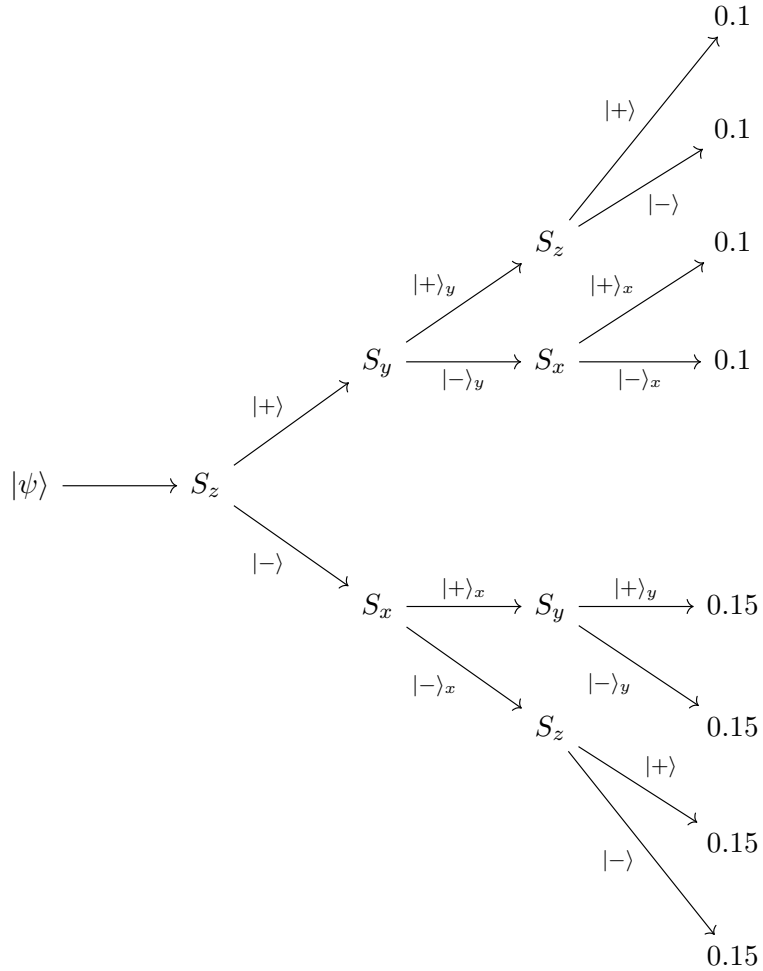
$$(A^2)^\dagger = \left(\begin{pmatrix} 1 & 2i \\ -2 & i \end{pmatrix} \begin{pmatrix} 1 & 2i \\ -2 & i \end{pmatrix} \right)^\dagger = \begin{pmatrix} 1-4i & 2i-2 \\ -2-2i & -4i-1 \end{pmatrix}^\dagger = \begin{pmatrix} 1+4i & -2+2i \\ -2-2i & -1+4i \end{pmatrix}.$$

22.

$$(AA^\dagger)^\dagger = \begin{pmatrix} 5 & \\ & 5 \end{pmatrix}^\dagger = \begin{pmatrix} 5 & \\ & 5 \end{pmatrix}.$$

Problem 2

1. Since we know the state after the first measurement, we should know all subsequent probabilities. We have shown multiple times that measurements of a basis state in an orthogonal basis yield probabilities of $1/2$ for each state, so we have that every measurement that originated at $S_z \rightarrow |+\rangle$ should be the same. Using this fact, we find that the measurements for $S_z \rightarrow |-\rangle$ have 0.15 probability, to ensure the sum of the probabilities is 1.



2. First we find the norm of the given vector.

$$3|+\rangle_x + 5i|-\rangle_x + 7|+\rangle_y \cong \frac{3}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} + \frac{5i}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} + \frac{7}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 10 + 5i \\ 3 + 2i \end{pmatrix}.$$

$$\| |\psi\rangle \| = \frac{1}{\sqrt{2}} \sqrt{|10 + 5i|^2 + |3 + 2i|^2} = \frac{1}{\sqrt{2}} \sqrt{125 + 13} = \frac{\sqrt{138}}{\sqrt{2}} = \sqrt{69}.$$

It follows that

$$|\psi\rangle = \frac{1}{\sqrt{69}} (3|+\rangle_x + 5i|-\rangle_x + 7|+\rangle_y).$$

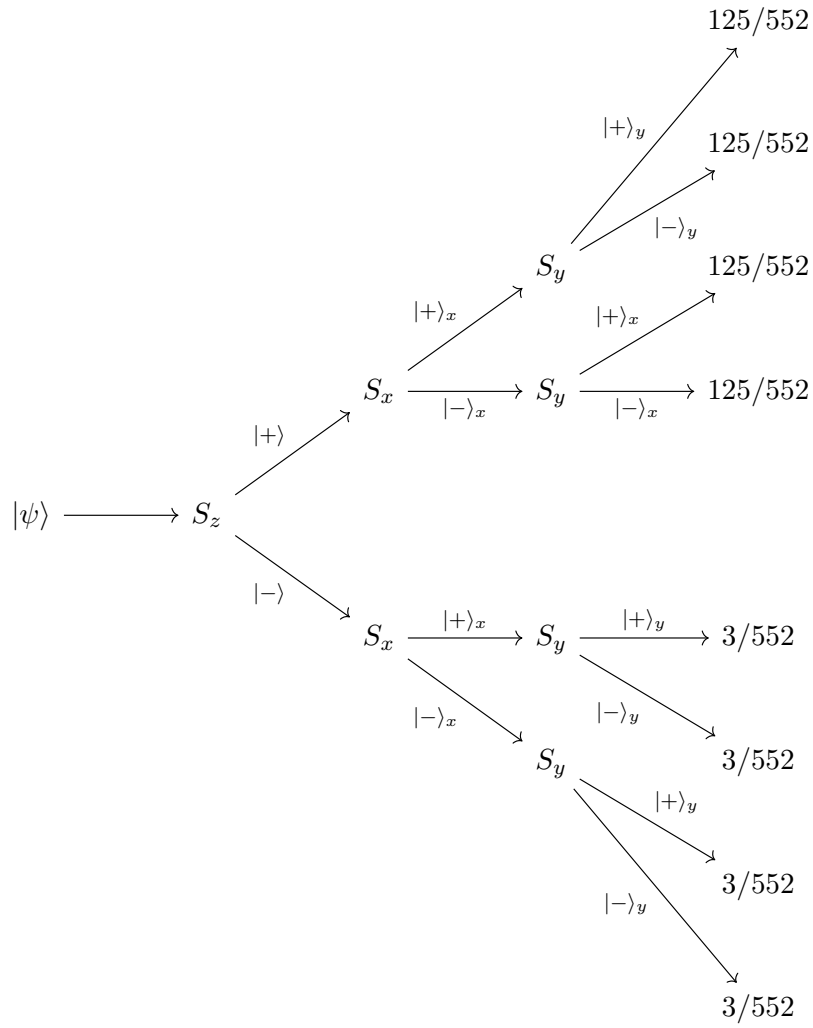
3. First, we compute the probabilities after the Z Stern-Gerlach machine. Note that

$$|\psi\rangle = \frac{10 + 5i}{\sqrt{138}}|+\rangle + \frac{3 + 2i}{\sqrt{138}}|-\rangle.$$

We have

$$\begin{aligned} \mathcal{P}_+ &= |\langle + | P_+ | \psi \rangle|^2 \\ &= |\langle + | + \rangle \langle + | \psi \rangle|^2 \\ &= |\langle + | \psi \rangle|^2 \\ &= \left| \frac{1}{\sqrt{138}} ((10 + 5i)\langle + | + \rangle + (3 + 2i)\langle + | - \rangle) \right|^2 \\ &= \left| \frac{10 + 5i}{\sqrt{138}} \right|^2 \\ &= \frac{125}{138}. \end{aligned}$$

After the first measurement, the distribution (inside only the top half or bottom half) will be even, since all subsequent measurements have 50/50 probabilities that we have proven enough times in class that I don't feel I need to do it again here. This means to get each final probability, just multiply the denominator by 4 to obtain 125/552. The tree is on the next page because it was too big to fit on this one. Also we clearly have $\mathcal{P}_- = 13/138$ to make the probabilities add to 1, so the other probabilities will be 13/552.



Problem 3

Given a unit vector of the form $\hat{n} = (\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta)$, the spin matrix $S_{\hat{n}}$ is given as

$$S_{\hat{n}} = \frac{\hbar}{2} \begin{pmatrix} \cos \theta & \sin \theta e^{-i\phi} \\ \sin \theta e^{i\phi} & -\cos \theta \end{pmatrix}.$$

In the problem, $\vec{n} = (\sin \theta, 0, \cos \theta)$ is of this form given that $\phi = 0$. We thus have

$$S_{\vec{n}} = \frac{\hbar}{2} \begin{pmatrix} \cos \theta & \sin \theta \\ \sin \theta & -\cos \theta \end{pmatrix}.$$

We also have

$$\begin{aligned}
S_{\vec{n}}|\psi_1\rangle &\cong \frac{\hbar}{2} \begin{pmatrix} \cos \theta & \sin \theta \\ \sin \theta & -\cos \theta \end{pmatrix} \begin{pmatrix} \cos(\theta/2) \\ \sin(\theta/2) \end{pmatrix} \\
&= \frac{\hbar}{2} \begin{pmatrix} \cos \theta \cos(\theta/2) + \sin \theta \sin(\theta/2) \\ \sin \theta \cos(\theta/2) - \cos \theta \sin(\theta/2) \end{pmatrix} \\
&= \frac{\hbar}{2} \begin{pmatrix} \cos \theta \cos(\theta/2) + 2 \sin^2(\theta/2) \cos(\theta/2) \\ 2 \sin(\theta/2) \cos^2(\theta/2) - \cos \theta \sin(\theta/2) \end{pmatrix} \\
&= \frac{\hbar}{2} \begin{pmatrix} (2 \cos^2(\theta/2) - 1) \cos(\theta/2) + 2 \sin^2(\theta/2) \cos(\theta/2) \\ 2 \sin(\theta/2) \cos^2(\theta/2) - (2 \cos^2(\theta/2) - 1) \sin(\theta/2) \end{pmatrix} \\
&= \frac{\hbar}{2} \begin{pmatrix} 2 (\cos^2(\theta/2) + \sin^2(\theta/2)) \cos(\theta/2) - \cos(\theta/2) \\ 2 \sin(\theta/2) \cos^2(\theta/2) - (2 \cos^2(\theta/2) \sin(\theta/2) - \sin(\theta/2)) \end{pmatrix} \\
&= \frac{\hbar}{2} \begin{pmatrix} \cos(\theta/2) \\ \cos(\theta/2) \sin(\theta/2) + \sin(\theta/2) \end{pmatrix} \\
&= \frac{\hbar}{2} \begin{pmatrix} \cos(\theta/2) \\ \sin(\theta/2) \end{pmatrix} \\
&\cong \frac{\hbar}{2} |\psi_1\rangle.
\end{aligned}$$

$$\begin{aligned}
S_{\vec{n}}|\psi_2\rangle &\cong \frac{\hbar}{2} \begin{pmatrix} \cos \theta & \sin \theta \\ \sin \theta & -\cos \theta \end{pmatrix} \begin{pmatrix} \sin(\theta/2) \\ -\cos(\theta/2) \end{pmatrix} \\
&= \frac{\hbar}{2} \begin{pmatrix} \cos \theta \sin(\theta/2) - \sin \theta \cos(\theta/2) \\ \sin \theta \sin(\theta/2) + \cos \theta \cos(\theta/2) \end{pmatrix} \\
&= \frac{\hbar}{2} \begin{pmatrix} -(\sin \theta \cos(\theta/2) - \cos \theta \sin(\theta/2)) \\ \sin \theta \sin(\theta/2) + \cos \theta \cos(\theta/2) \end{pmatrix} \\
&= \frac{\hbar}{2} \begin{pmatrix} -\sin(\theta/2) \\ \cos(\theta/2) \end{pmatrix} \\
&\cong -\frac{\hbar}{2} |\psi_2\rangle.
\end{aligned}$$

Here I've used the fact that the top entry of $S_{\vec{n}}|\psi_2\rangle$ is the negative of the bottom entry of $S_{\vec{n}}|\psi_1\rangle$, so we can just plug in the answer from there, and similarly the bottom entry of $S_{\vec{n}}|\psi_2\rangle$ is the top entry of $S_{\vec{n}}|\psi_1\rangle$.

Consider $S_{\vec{n}}^2$.

$$\frac{\hbar}{2} \begin{pmatrix} \cos \theta & \sin \theta \\ \sin \theta & -\cos \theta \end{pmatrix} \frac{\hbar}{2} \begin{pmatrix} \cos \theta & \sin \theta \\ \sin \theta & -\cos \theta \end{pmatrix} = \frac{\hbar^2}{4} \begin{pmatrix} \cos^2 \theta + \sin^2 \theta & \cos \theta \sin \theta - \sin \theta \cos \theta \\ \sin \theta \cos \theta - \cos \theta \sin \theta & \sin^2 \theta + \cos^2 \theta \end{pmatrix} = \frac{\hbar^2}{4} \mathbb{1}.$$

Therefore, $S_{\vec{n}}^2 = (\hbar^2/4) \mathbb{1}$.